

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-004: Student Input Field Validation for Add and Edit Operations

Field	Details
Change Request ID	CR-004
Change Request Title	Student Input Field Validation for Add and Edit Operations
Student Name	J.Thushany
Student ID	JF24ICT09
Implementation Status	Completed
Date	8.10.2026

1. Change Request Summary

1.1 Business Requirement

Administrators previously could add and edit student records with insufficient input validation. Invalid registration numbers, short names, invalid characters, incorrect email addresses, invalid phone numbers, and weak passwords could be entered.

The change was required to introduce proper validation for student Add and Edit operations at both frontend and backend levels. The validation must prevent invalid student information from being submitted and stored while preserving the existing student management functionality.

1.2 Existing Behavior

Before implementation:

- The administrator dashboard provided Add and Edit Student functionality.
- Basic required-field validation was available in the student form.
- First and last names did not have sufficient length and character validation.
- Email validation was insufficient.
- Phone numbers were not comprehensively validated.
- New student passwords did not enforce a complete password-strength policy.
- Backend validation did not comprehensively validate all student input fields.
- Existing Edit and Delete operations used the existing student identifiers.

1.3 Requested Change

Input validation was required for student Add and Edit operations.

The validation must:

- Validate registration number before submission.
- Require first and last names to contain at least 3 characters.
- Allow alphabetic characters in names while rejecting numbers and inappropriate special characters.
- Validate email addresses using a valid email format.
- Require phone numbers to contain exactly 10 numeric digits.
- Require a strong password when creating a new student.
- Require passwords to contain at least 8 characters, uppercase letters, lowercase letters, numbers, and special characters.
- Display clear field-specific validation error messages.
- Prevent invalid frontend submissions.
- Apply server-side validation to prevent validation bypass.
- Preserve existing Add, Edit, Delete, authentication, authorization, and password hashing functionality.

2. Implementation

2.1 Implementation Summary

The CR-004 requirements were implemented in the student management functionality.

Client-side validation was added to the Student Add/Edit form. The form validates registration number, first name, last name, email, phone number, and password before submitting the request.

Server-side validation was also implemented in the student controller to prevent invalid data from bypassing frontend validation.

Field-specific validation messages are displayed below the relevant input field. Invalid submissions are stopped before the API request is sent.

The existing Add, Edit, Delete, authentication, authorization, and password hashing functionality was preserved.

2.2 Affected Files

File	Change	Purpose
frontend/src/components/StudentFormModal.jsx	Modified	Implemented client-side validation for registration number, names, email, phone number, and password, including field-specific error messages.
frontend/src/components/StudentFormModal.css	Added	Provides styling for student form validation error messages.
backend/controllers/studentController.js	Modified	Implemented server-side validation and duplicate registration/email checks while preserving existing CRUD and password hashing functionality.

Additional files: None.

2.3 Technical Changes

Registration Number Validation

The registration number is validated before submission.

The validation:

- Rejects an empty registration number.
- Allows letters, numbers, hyphens, and slashes.
- Prevents invalid special characters.

First Name and Last Name Validation

The first name and last name are validated using the following rules:

- Field cannot be empty.
- Minimum length is 3 characters.
- Alphabetic characters are accepted.
- Numbers and inappropriate special characters are rejected.
- Spaces, apostrophes, and hyphens can be used where appropriate.

Examples of validation messages include:

- First name must contain at least 3 characters.
- Last name must contain at least 3 characters.
- First name can contain only alphabetic characters.
- Last name can contain only alphabetic characters.

Email Validation

Email input is validated using a valid email format.

Invalid values such as incomplete email addresses are rejected before submission.

Example validation message:

- Please enter a valid email address.

Phone Number Validation

Phone numbers are validated using the following rules:

- Phone number is required.
- Only numeric characters are accepted.
- Exactly 10 digits are required.

Example validation messages include:

- Phone number must contain numeric characters only.
- Phone number must contain exactly 10 digits.

Password Validation

For new student creation, the password is validated using a strong password policy.

The password must:

- Contain at least 8 characters.
- Contain at least one uppercase letter.
- Contain at least one lowercase letter.
- Contain at least one number.
- Contain at least one special character.

Password validation is required during student creation. Existing password hashing using bcrypt remains unchanged.

2.4 Implementation Logic

1. Administrator opens the Add Student or Edit Student form.

2. Existing student information is loaded when editing a student.
3. Administrator enters or modifies student information.
4. The frontend validates all applicable input fields.
5. Field-specific validation errors are displayed for invalid fields.
6. Submission is prevented when validation errors exist.
7. Valid input is submitted to the existing student API endpoint.
8. The backend validates the received student information again.
9. Invalid API requests are rejected by server-side validation.
10. Valid student information is saved using the existing CRUD functionality.
11. New student passwords continue to be securely hashed using bcrypt.
12. Existing authentication and authorization controls remain unchanged.

2.5 Error and Data Handling

Client-side validation errors are displayed next to the relevant input field.

Invalid form submissions are prevented before the API request is generated.

Server-side validation provides an additional protection layer against invalid or directly submitted API requests.

The implementation also checks for duplicate registration numbers and email addresses during student creation/update.

Validation errors do not expose sensitive information.

Existing authentication and authorization error handling remains unchanged.

3. Testing & Verification

3.1 Testing Approach

The implementation was verified through manual frontend testing.

Testing Types:

- Functional testing
- Positive testing
- Validation testing

- UI testing
- Regression testing
- Error handling testing
- CRUD verification
- Server-side validation verification

Environment: Development

Browser: Chrome

Test Role: Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Submit empty required fields	Required-field validation messages are displayed	Required-field validation displayed	Pass
TC-02	Enter first name with fewer than 3 characters	First name validation message is displayed	"First name must contain at least 3 characters." displayed	Pass
TC-03	Enter last name with fewer than 3 characters	Last name validation message is displayed	"Last name must contain at least 3 characters." displayed	Pass
TC-04	Enter invalid characters in first name	Invalid character validation message is displayed	"First name can contain only alphabetic characters." displayed	Pass
TC-05	Enter invalid email address	Email validation message is displayed	"Please enter a valid email address." displayed	Pass
TC-06	Enter phone number with fewer than 10 digits	Phone length validation message is displayed	"Phone number must contain exactly 10 digits." displayed	Pass
TC-07	Enter non-numeric phone number	Numeric-only validation message is displayed	"Phone number must contain numeric characters only." displayed	Pass
TC-08	Enter a valid 10-digit phone number	Phone number is accepted	Valid phone number accepted	Pass

TC-09	Enter weak password during student creation	Password-strength validation message is displayed	Weak password rejected	Pass
TC-10	Enter strong password during student creation	Password is accepted	Strong password accepted	Pass
TC-11	Submit valid student information	Student is created successfully	Student record created successfully	Pass
TC-12	Submit invalid student information	Student is not created/updated	Invalid submission prevented	Pass
TC-13	Edit student using invalid information	Update is prevented	Invalid update prevented	Pass
TC-14	Edit student using valid information	Student information is updated	Valid update accepted	Pass
TC-15	Verify Edit and Delete functionality	Correct student record is affected	Existing CRUD functionality worked correctly	Pass
TC-16	Verify server-side validation	Invalid direct API input is rejected	Server-side validation implemented	Pass

3.3 Test Summary

Test Result	Count
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Total Test Cases 16

Passed	16
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Failed	0
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Pass Rate	100%
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Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Empty required fields are rejected.	Empty-field validation test	Met
AC-002	First names shorter than 3 characters are rejected.	First name validation test	Met
AC-003	Last names shorter than 3 characters are rejected.	Last name validation test	Met
AC-004	Invalid characters in names are rejected.	Invalid character test	Met
AC-005	Invalid email addresses are rejected.	Email validation test	Met
AC-006	Phone numbers shorter than 10 digits are rejected.	Phone length test	Met
AC-007	Phone numbers longer than 10 digits are rejected.	Phone length validation	Met
AC-008	Non-numeric phone numbers are rejected.	Phone character test	Met
AC-009	Valid 10-digit phone numbers are accepted.	Valid phone test	Met
AC-010	Weak passwords are rejected.	Password validation test	Met
AC-011	Strong passwords are accepted.	Strong password test	Met
AC-012	Valid student information can be created.	Valid Add Student test	Met
AC-013	Invalid student information cannot be created.	Invalid Add Student test	Met
AC-014	Invalid student information cannot be used for update.	Invalid Edit Student test	Met

AC-015	Valid student information can be updated.	Valid Edit Student test	Met
AC-016	Direct API invalid input is rejected by server-side validation.	Backend validation verification	Met
AC-017	Existing Edit and Delete functionality continues to work.	CRUD regression test	Met
AC-018	Existing authentication and authorization remain unchanged.	Authorization regression verification	Met

5. Evidence

Frontend Validation Evidence

Evidence-01 – First Name Minimum Length Validation

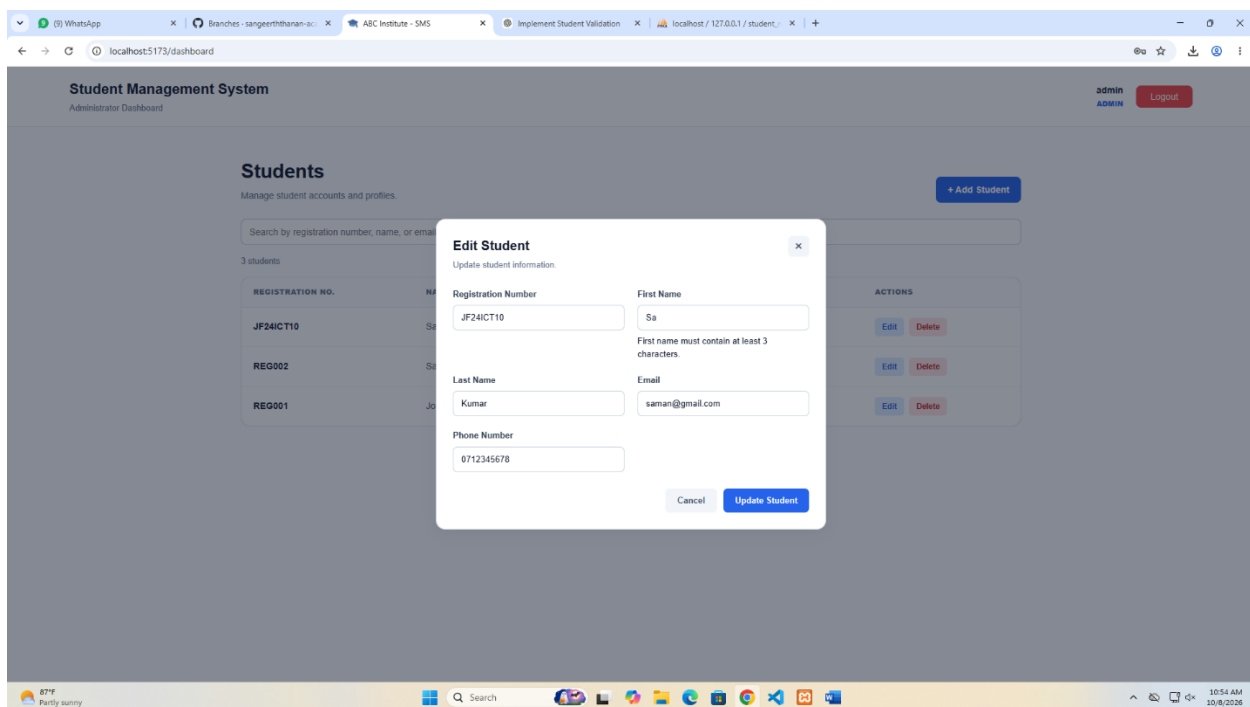


Figure 1: Edit Student form displaying a validation error when the first name contains fewer than 3 characters.

Evidence-02 – Last Name Minimum Length Validation

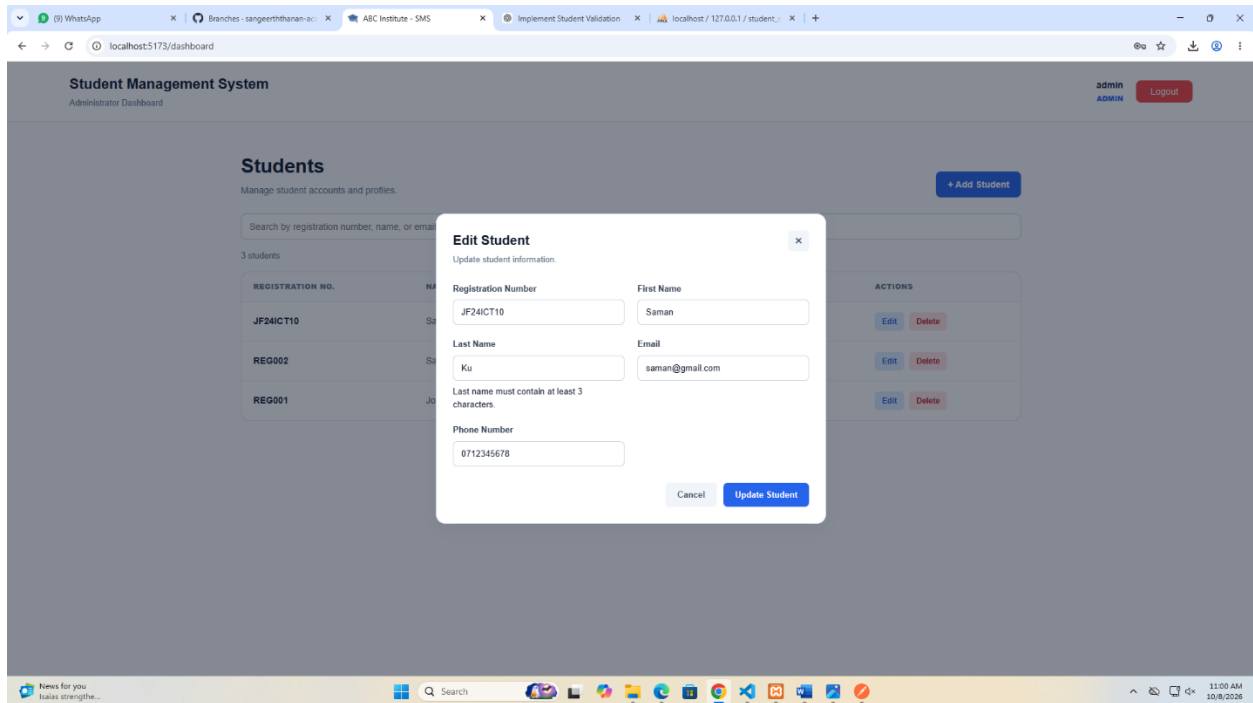


Figure 2: Edit Student form displaying a validation error when the last name contains fewer than 3 characters.

Evidence-03 – Invalid First Name Character Validation

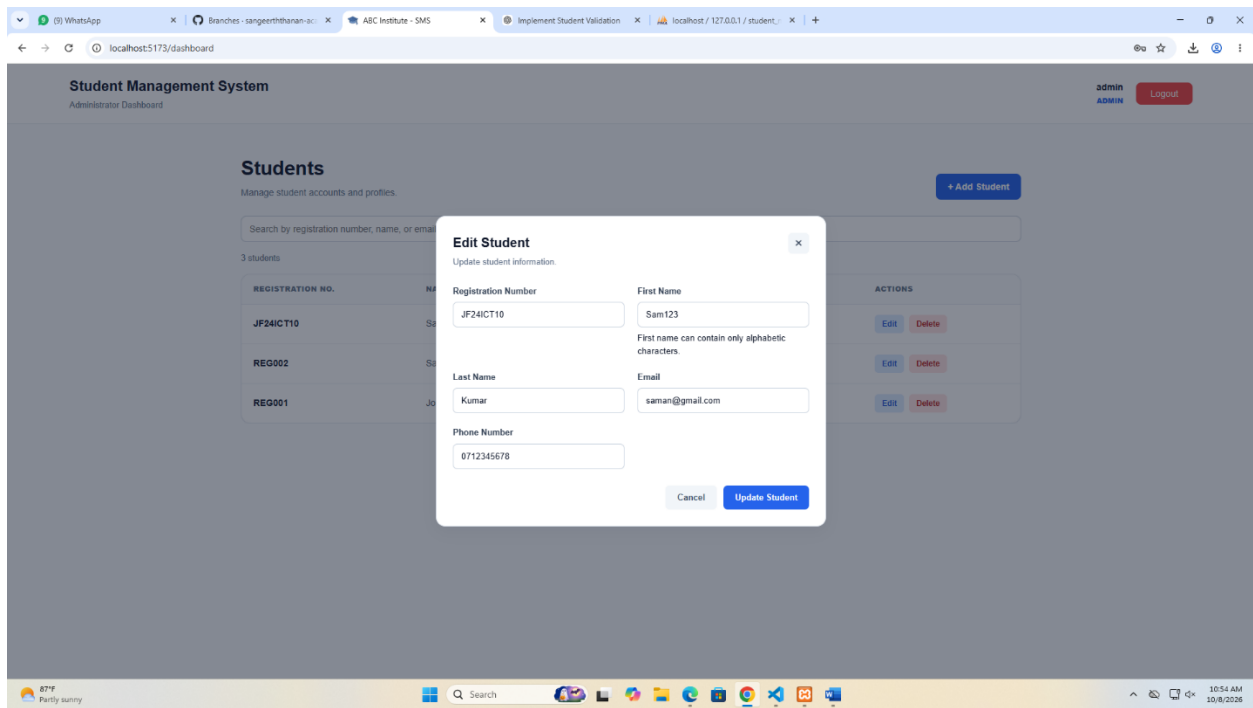


Figure 3: Edit Student form displaying a validation error when the first name contains numeric characters.

Evidence-04 – Invalid Email Validation

The screenshot shows a web browser window with the 'Student Management System' dashboard. A modal titled 'Edit Student' is open, displaying a form to update student information. The form includes fields for Registration Number, First Name, Last Name, Email, and Phone Number. The Email field contains 'test@' and a red validation error message 'Please enter a valid email address.' is displayed below it. The First Name field contains 'Saman'. The Registration Number field contains 'JF24ICT10'. The Last Name field contains 'Kumar'. The Phone Number field contains '0712345678'. The modal has 'Cancel' and 'Update Student' buttons. In the background, a table lists students with columns for Registration No., Name, and Actions (Edit, Delete).

REGISTRATION NO.	NAME	ACTIONS
JF24ICT10	Saman	Edit Delete
REG002	Saman	Edit Delete
REG001	John	Edit Delete

Figure 4: Edit Student form displaying a validation error for an invalid email address.

Evidence-05 – Phone Number Length Validation

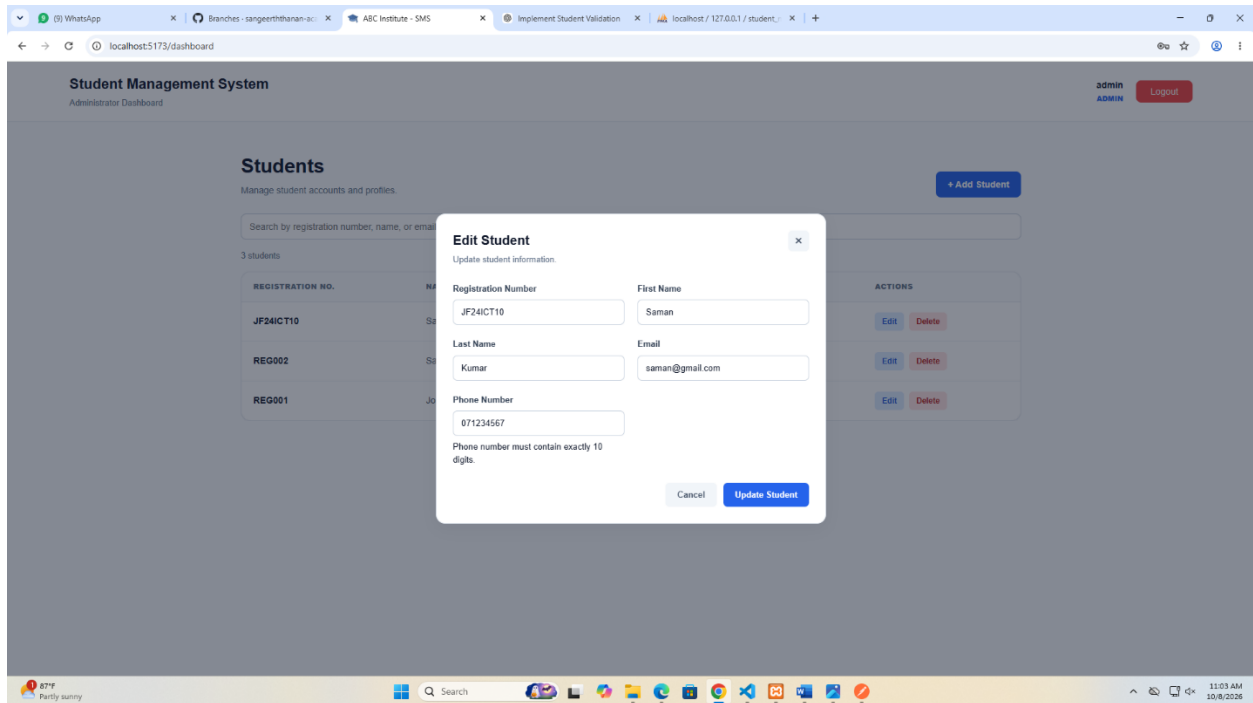


Figure 5: Edit Student form displaying a validation error when the phone number contains fewer than 10 digits.

Evidence-06 – Phone Number Character Validation

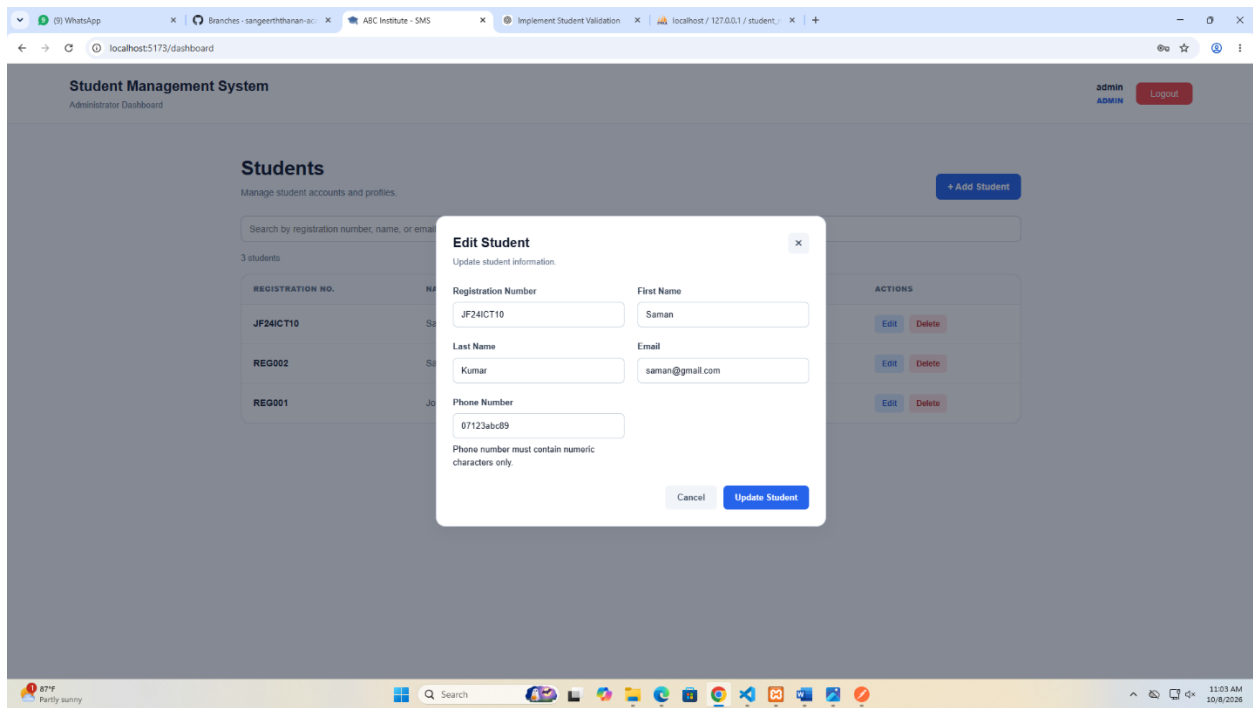


Figure 6: Edit Student form displaying a validation error when the phone number contains non-numeric characters.

Backend Validation Evidence

Evidence-07 – Server-Side First Name Validation

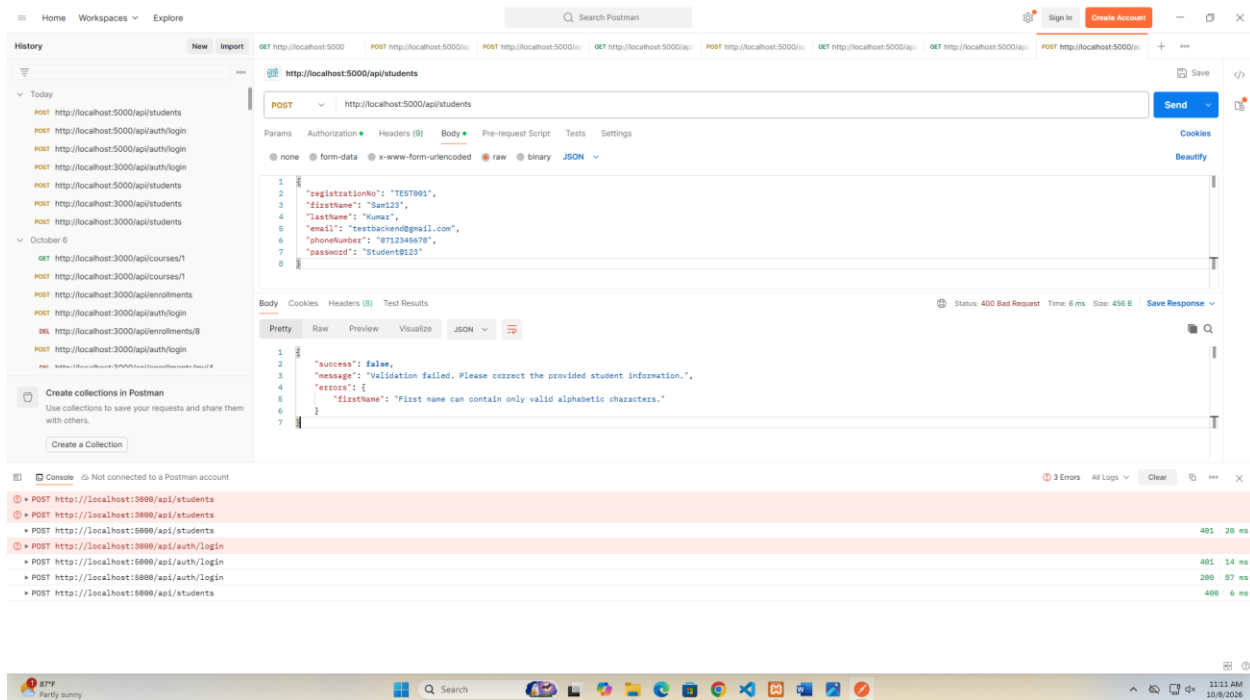


Figure 7: Postman response demonstrating server-side validation of an invalid first name. The backend rejects the student submission when the first name contains numeric characters and returns a field-specific validation error.

Evidence-08 – Server-Side Email Validation

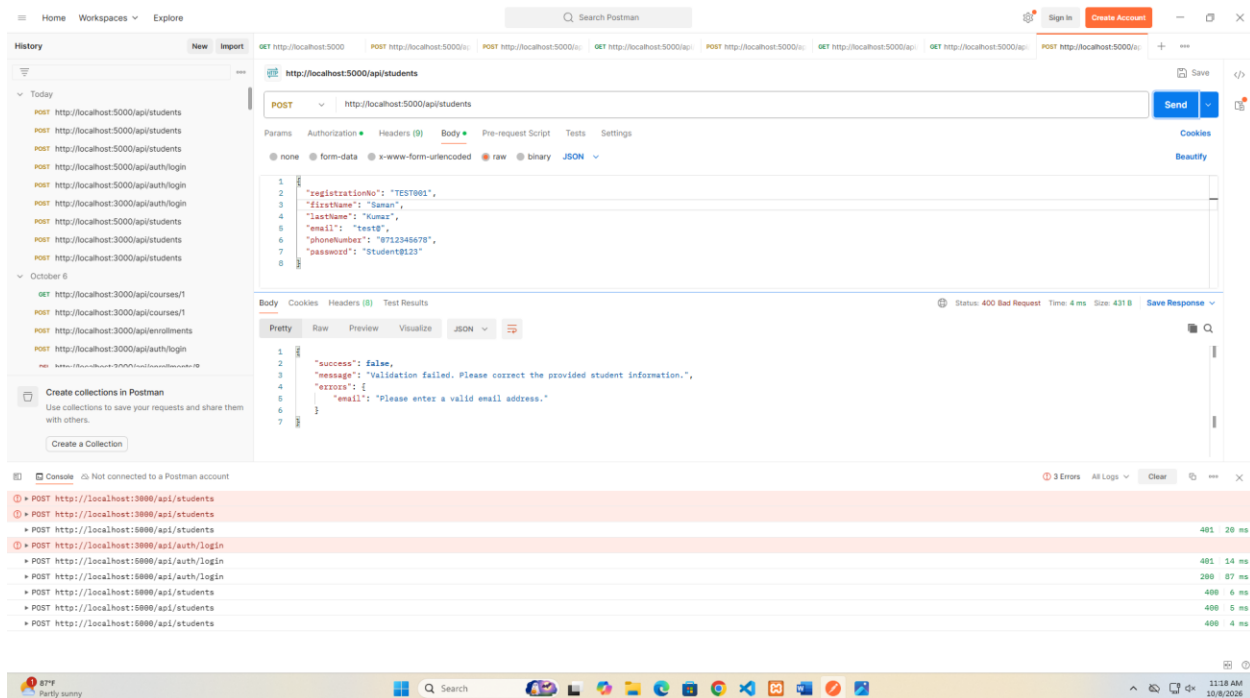


Figure 8: Postman response demonstrating server-side validation of an invalid email address. The backend rejects the student submission and returns a field-specific validation error for email.

Evidence-09 – Server-Side Phone Number Validation

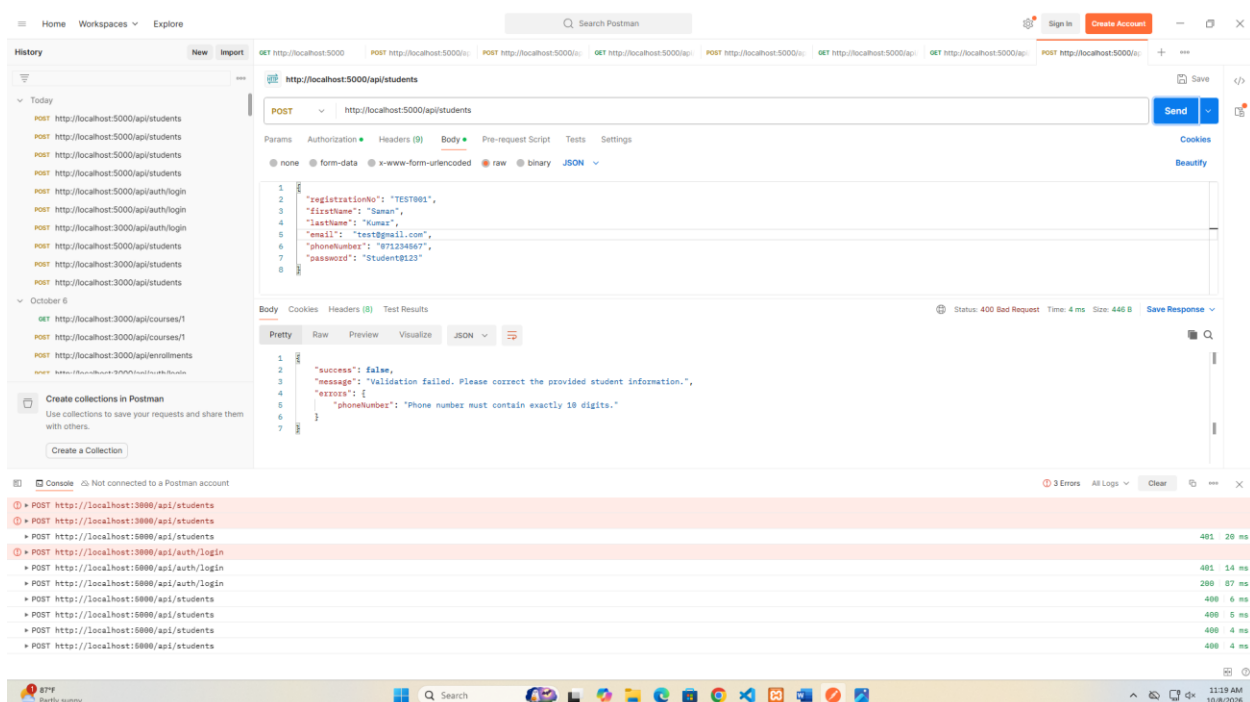


Figure 9: Postman response demonstrating server-side validation when the phone number contains fewer than 10 digits. The backend rejects the student submission and returns a field-specific validation error for phone Number.

Evidence-10 – Server-Side Password Validation

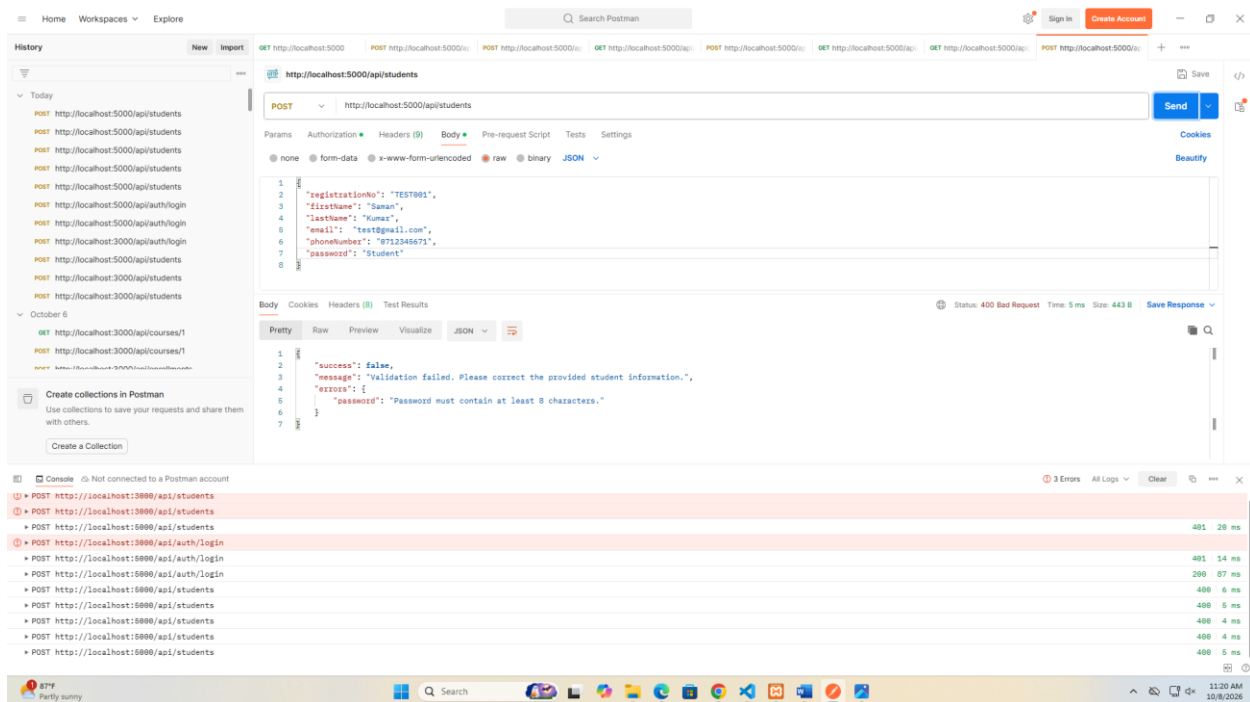


Figure 10: Postman response demonstrating server-side validation of a weak password (password). The backend rejects the student submission and returns a field-specific validation error for password.

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing CRUD	Verified
Final Status	Accepted

6.1 Final Summary

CR-004 was implemented as a student input validation enhancement for Add and Edit operations in the administrator student management functionality.

The implementation validates registration number, first name, last name, email address, phone number, and new student passwords. Client-side validation prevents invalid submissions and provides clear field-specific error messages.

Server-side validation was also implemented to prevent invalid input from bypassing frontend validation. Password hashing, authentication, authorization, and existing Add, Edit, and Delete functionality were preserved.

The implemented functionality was manually tested against the defined validation, functional, security, and acceptance requirements. All tested scenarios passed successfully.

6.2 Issues / Limitations

No significant issues identified.

Student Name: _____

Student ID: _____

Date: 8 October 2026

Signature: _____

END OF REPORT